

Data Structures and Algorithms

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Objectives

Minimum Spanning Tree

Kruskal's Algorithm

Analysis of Kruskal's Algorithm

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Minimum Spanning Tree

Spanning Tree

Given a graph G=(V,E), a spanning tree H=(S,T) is an acyclic subgraph that contains all the vertices(V=S), and a subset of edges (T⊆E) of G.

A spanning tree of an undirected graph is a tree formed by graph edges that connect all the vertices of the graph.

Minimum Spanning Tree

The minimum spanning tree is a connected subgraph of G that spans all vertices at minimum cost.

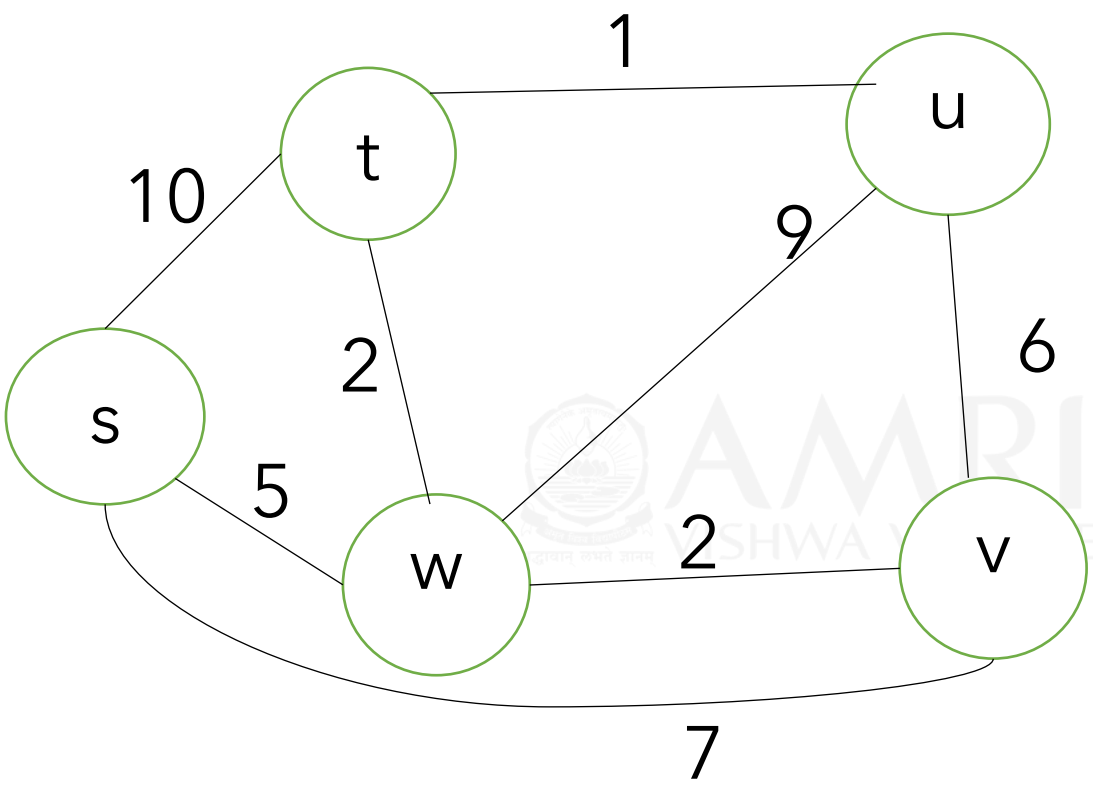
Minimum Spanning Tree

```
1  Kruskal(G, w):
2      A = ∅
3      for each vertex v ∈ G.V do
4          Make-Set(v)
5      end
6      sort the edges of G.E by non-decreasing order by weight w.
7      for each edge(u, v) ∈ G.E taken in ascending order by weight do
8          if Find − Set(u) ≠ Find − Set(v) then
9              |A| = A ∪ {(u, v)}
10             Union(u, v)
11         end
12     end
13     return A
14 end
```

Introduction to Algorithms, Third EditionSeptember 2009 ,Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest Clifford Stein



Kruskal's Algorithm



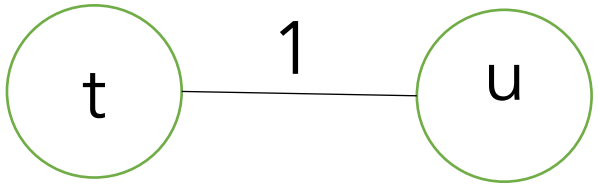
A={}

$\{s\}$ $\{t\}$ $\{w\}$ $\{u\}$ $\{v\}$

Sorted edges
(t,u).(t,w),(w,v),(s,w),(v,u),(v.s), (s,t)

Kruskal's Algorithm

Minimum Spanning Tree



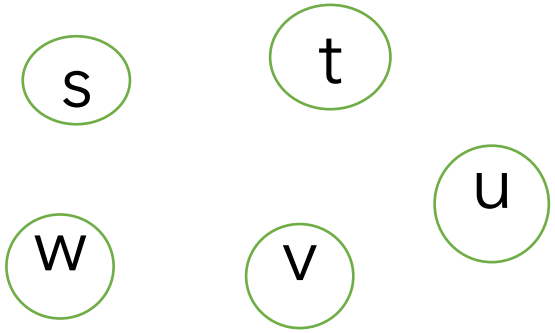
A={}

$\{s\}\ \{t\}\ \{w\}\ \{u\}\ \{v\}$

Sorted edges
(t,u),(t,w),(w,v),
(s,w),(v,u),(v.s),
(s,t)

Select (t,u)
Find-set(t) ≠ Find-set(u)
A = A U {(t,u)}
Union(t,u)

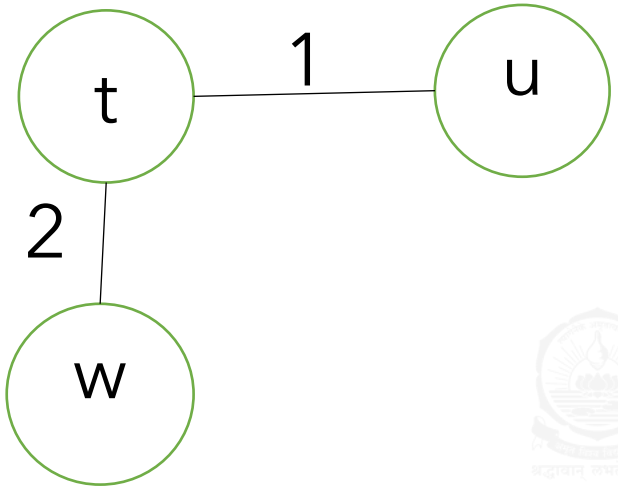
disjoint-set forest



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Kruskal's Algorithm

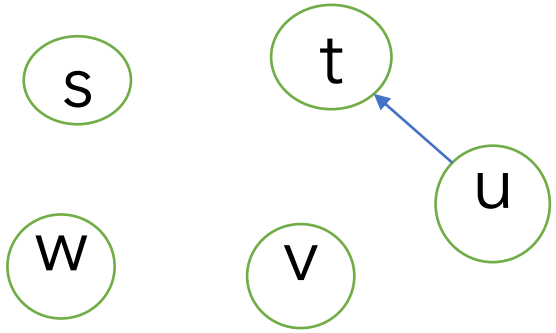
Minimum Spanning Tree



A={(t,u)}

$$\{s\}\ \{t,u\}\ \{w\}\ \{v\}$$

disjoint-set forest

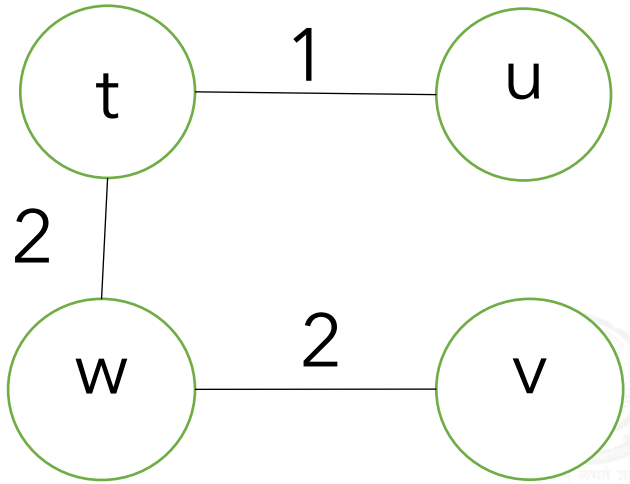


- Sorted edges (t,w),(w,v),(s,w),(v,u), (v.s), (s,t)
- Select (t,w)
- Find-set(t) ≠ Find-set(w)
- A = A U {(t,w)}
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Kruskal's Algorithm

Minimum Spanning Tree

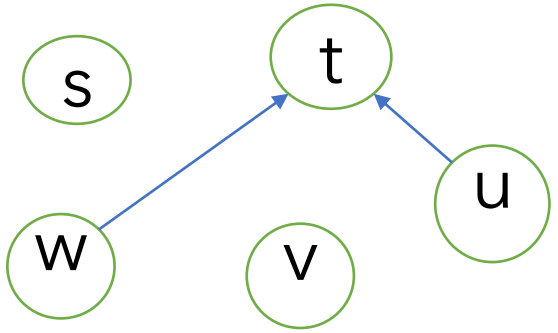


A={(t,u),(t,w)}

$\{s\}$ $\{t,u,w\}$ $\{v\}$

- Sorted edges (w,v),(s,w),(v,u),(v.s) , (s,t)
- Select (w,v)
- Find-set(w) ≠ Find-set(v)
- A = A U {(w,v)}
- Union(w,v)

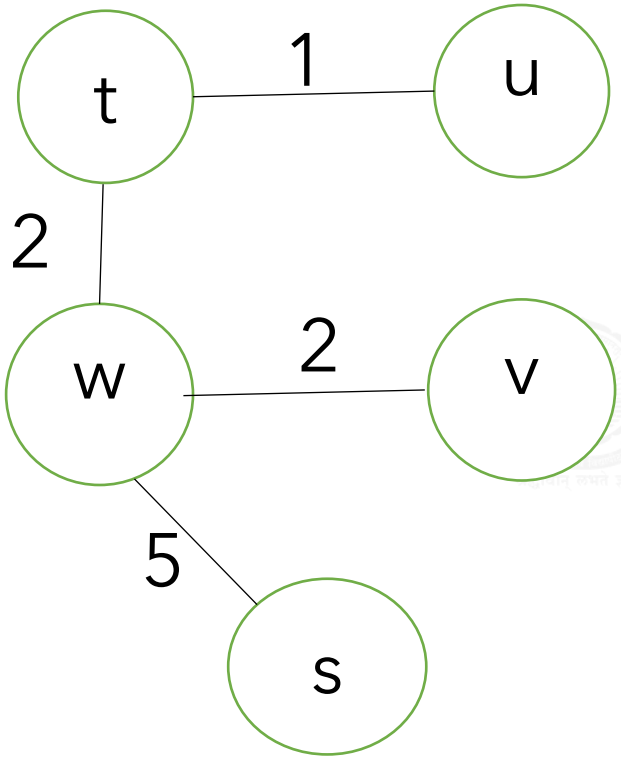
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Kruskal's Algorithm

Minimum Spanning Tree

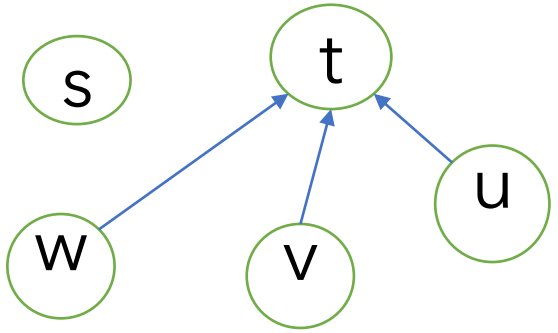


A={(t,u),(t,w),(w,v)}

$\{s\}$ $\{t, u, w, v\}$

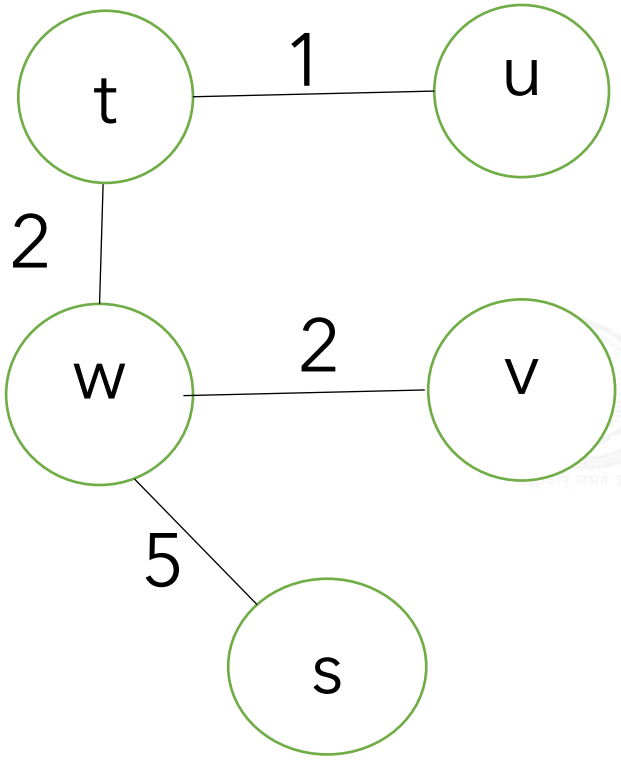
- Sorted edges (s,w),(v,u),(v.s), (s,t)
- Select (s,w)
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disjoint-set forest



Kruskal's Algorithm

Minimum Spanning Tree



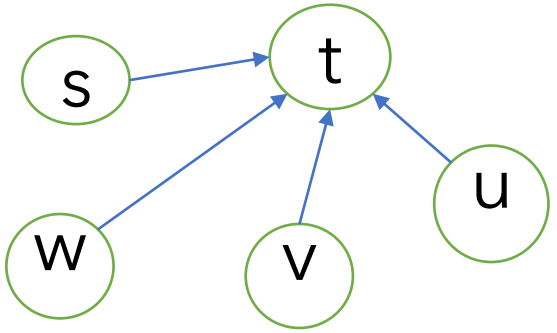
A={(t,u),(t,w),(w,v),(s,w)}

$\{s, t, u, w, v\}$

- Sorted edges (v,u),(v,s),(s,t)
- Select (v,u)
- Find-set(v) ≠ Find-set(u) →false since v and u belongs to same set.

- Minimum cost - 10

Tree- union of sets



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Kruskal's Algorithm - Analysis

```
1  Kruskal(G, w):
2      A = ∅
3      for each vertexv ∈ G.V do
4          Make-Set(v)
5      end
6      sort the edges of G.E by non-decreasing order by weight w.
7      for each edge(u, v) ∈ G.E taken in ascending order by weight do
8          if Find − Set(u) ≠ Find − Set(v) then
9              |A| = A ∪ {(u, v)}
10             Union(u, v)
11         end
12     end
13     return A
14 end
```

Line – 2 – O(1)
Line 3-4 O(V)
Line – 6 - O(E logE)
Line 7–O(E)
Line 8-10 O(log V)

Total running time-
O(V)+O(ElogE)+O(E*logV)
= **O(ElogE)**

Summary

Kruskal's algorithm – Greedy Algorithm

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References

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Objectives

Prim's Algorithm

Analysis of Prim's Algorithm

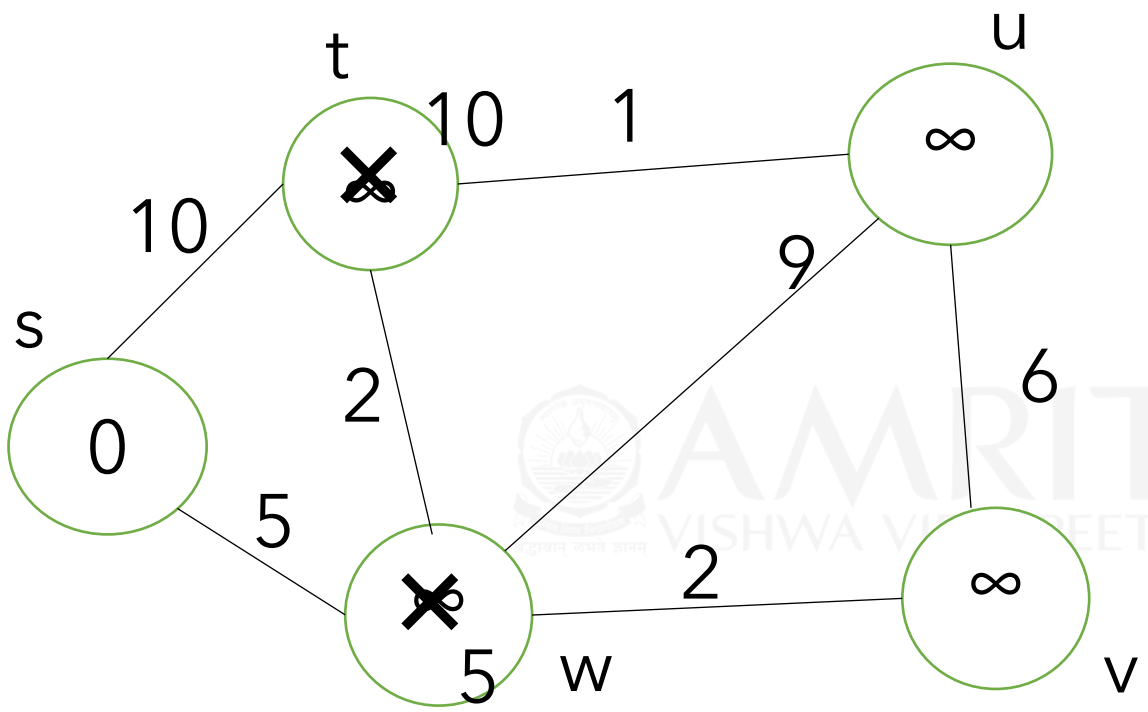
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Prim's Algorithm

```
1  PRIMS(G, w, s):
2      for each u ∈ G.V do
3          u.key = ∞
4          u.pred = NIL
5      end
6      s.key = 0
7      Q = G.V
8      while Q ≠ ∅ do
9          u = ExtractMin(Q)
10         for each v ∈ G.Adj[u] do
11             if v ∈ Q and w(u, v) < v.key then
12                 v.pred = u
13                 v.key = w(u, v)
14             end
15         end
16     end
17 end
```

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Prims's Algorithm

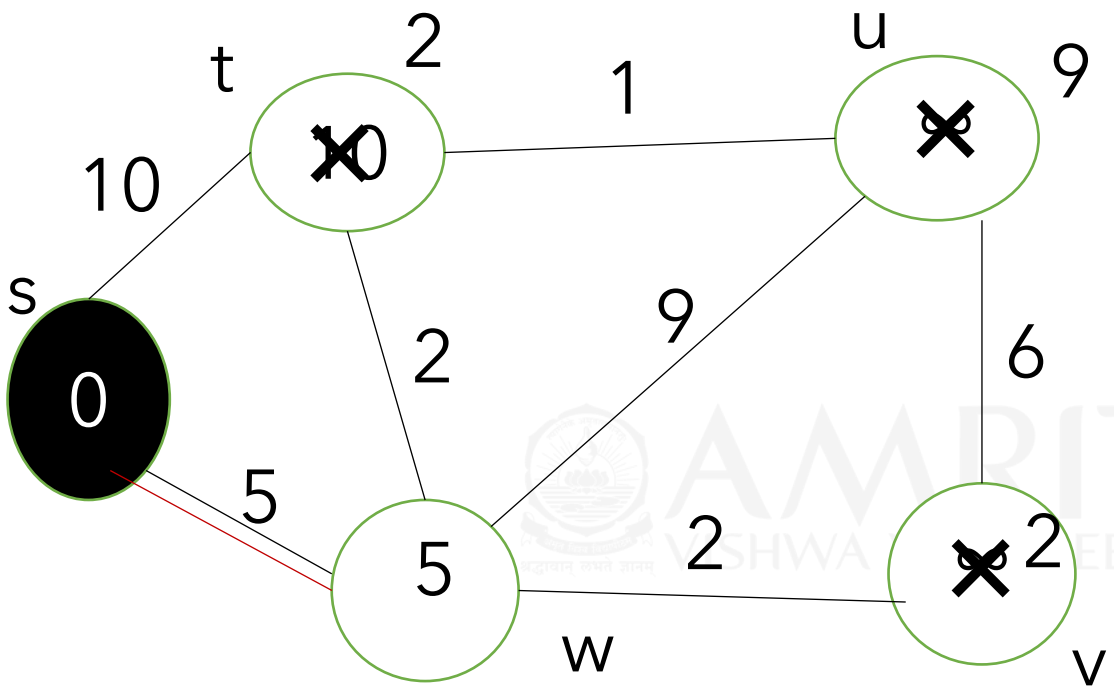


Priority Queue

~~∞~~	t	w	u	v

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Prims's Algorithm

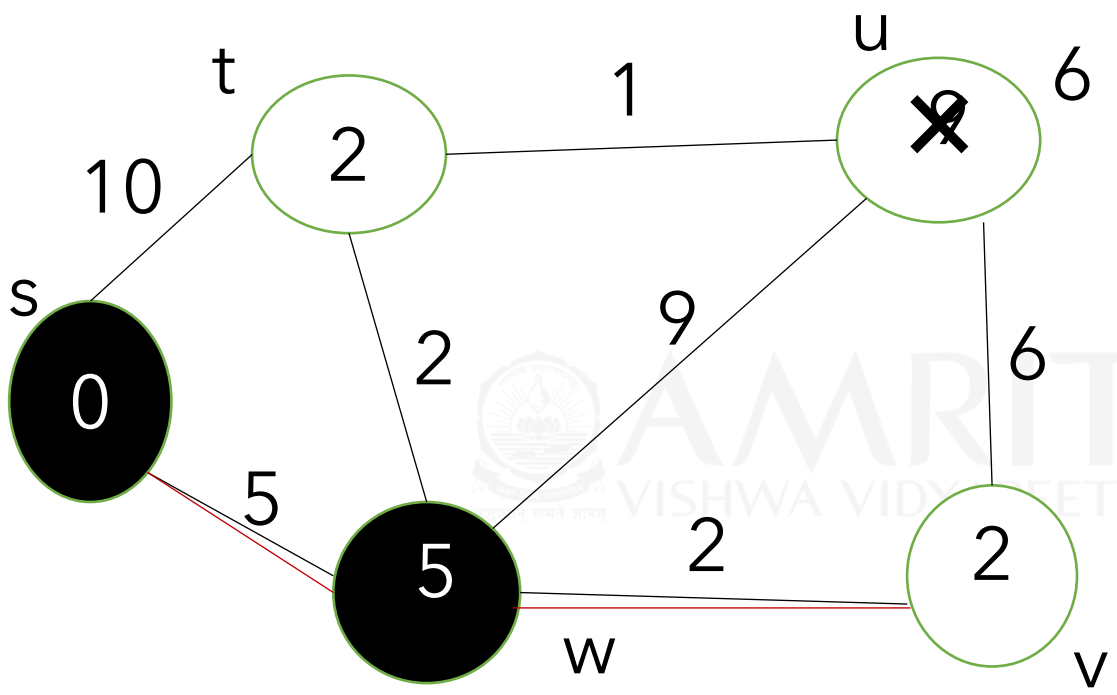


Priority Queue

	t	~~v~~	u	v

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Prims's Algorithm

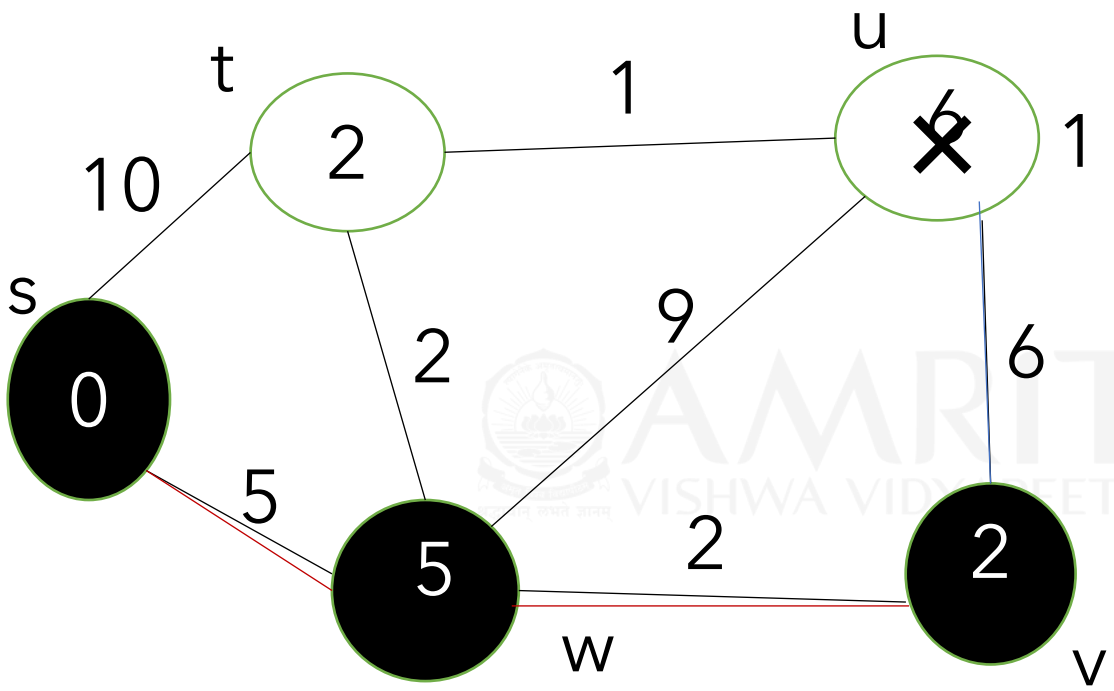


Priority Queue

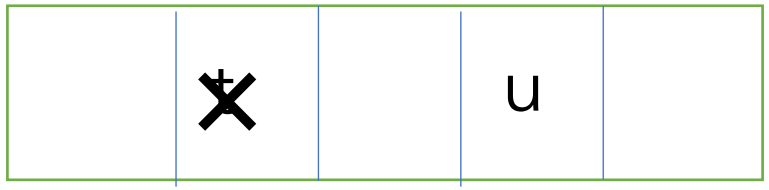
	t		u	~~x~~

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Prims's Algorithm

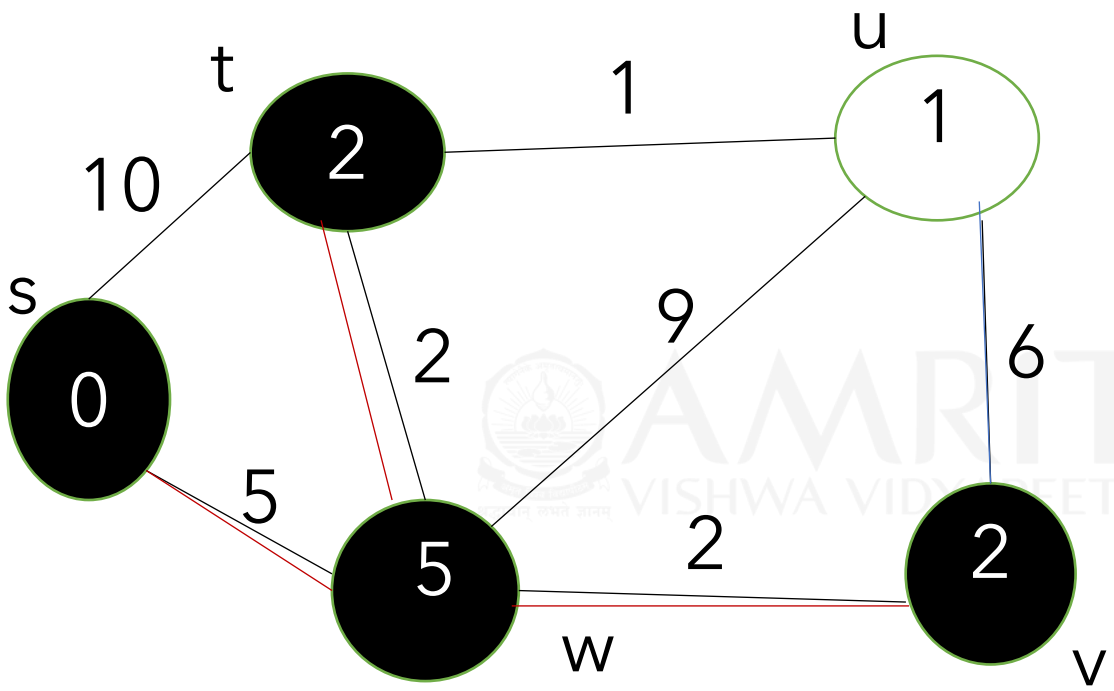


Priority Queue

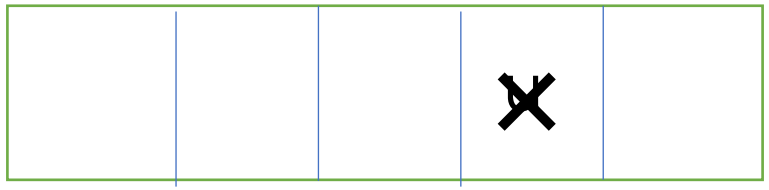


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Prims's Algorithm

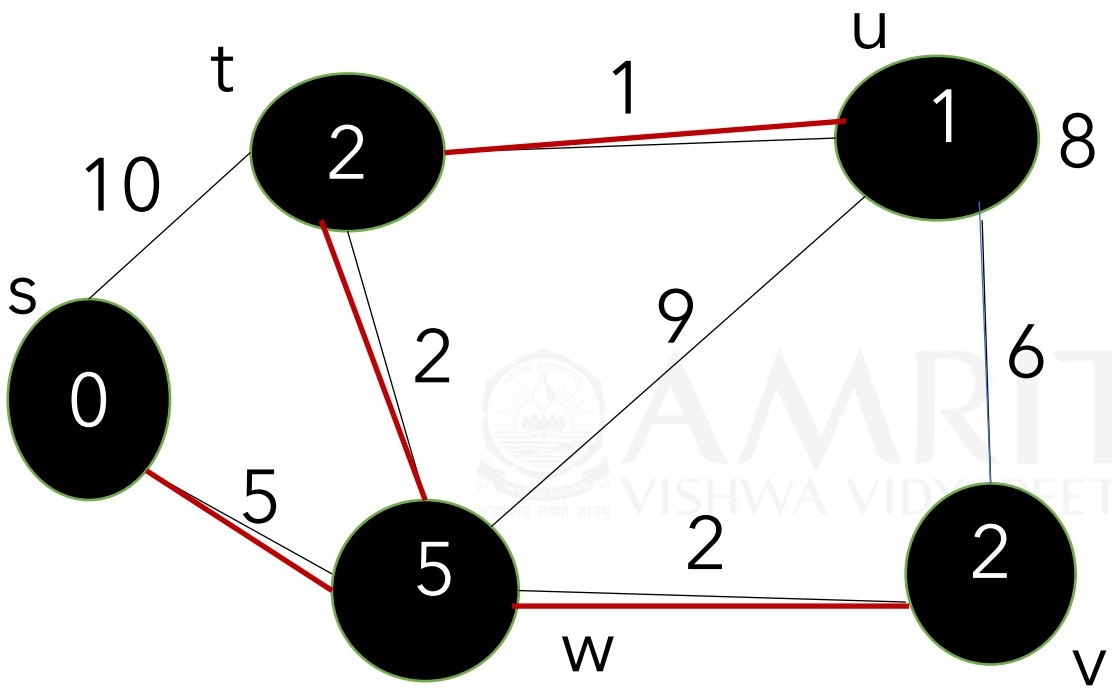


Priority Queue



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Prims's Algorithm



Edges marked in red form edges of the minimum spanning tree.

Cost=10

Analysis

```
1  PRIMS(G, w, s):
2      for each u ∈ G.V do
3          u.key = ∞
4          u.pred = NIL
5      end
6      s.key = 0
7      Q = G.V
8      while Q ≠ ∅ do
9          u = ExtractMin(Q)
10         for each v ∈ G.Adj[u] do
11             if v ∈ Q and w(u, v) < v.key then
12                 v.pred = u
13                 v.key = w(u, v)
14             end
15         end
16     end
17 end
```

If Queue is implemented as a binary heap then building a min-heap is O(V).

- Lines 2-5 – O(V)
- EXTRACT-MIN operation takes O(lgV) time, it executes |V| times, so the total time for all calls to EXTRACT-MIN is O(V lgV).
- For loop in lines 10-15 executes O(E), since the total length of adjacency lists of all nodes is 2*|E|.
- Assignment operation in line 13 takes O(logV) times.

- Total time for Prim's algorithm is O(V lg V + E lg V) =
- **O(E lgV)**

Summary

Prim's – greedy algorithm

Works with undirected connected graph

Prim's assumes all vertices are connected.

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References

Introduction to Algorithms – Thomas H Cormen

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Data Structures and Algorithms

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Objectives

Dijkstra's Algorithm

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Dijkstra's algorithm

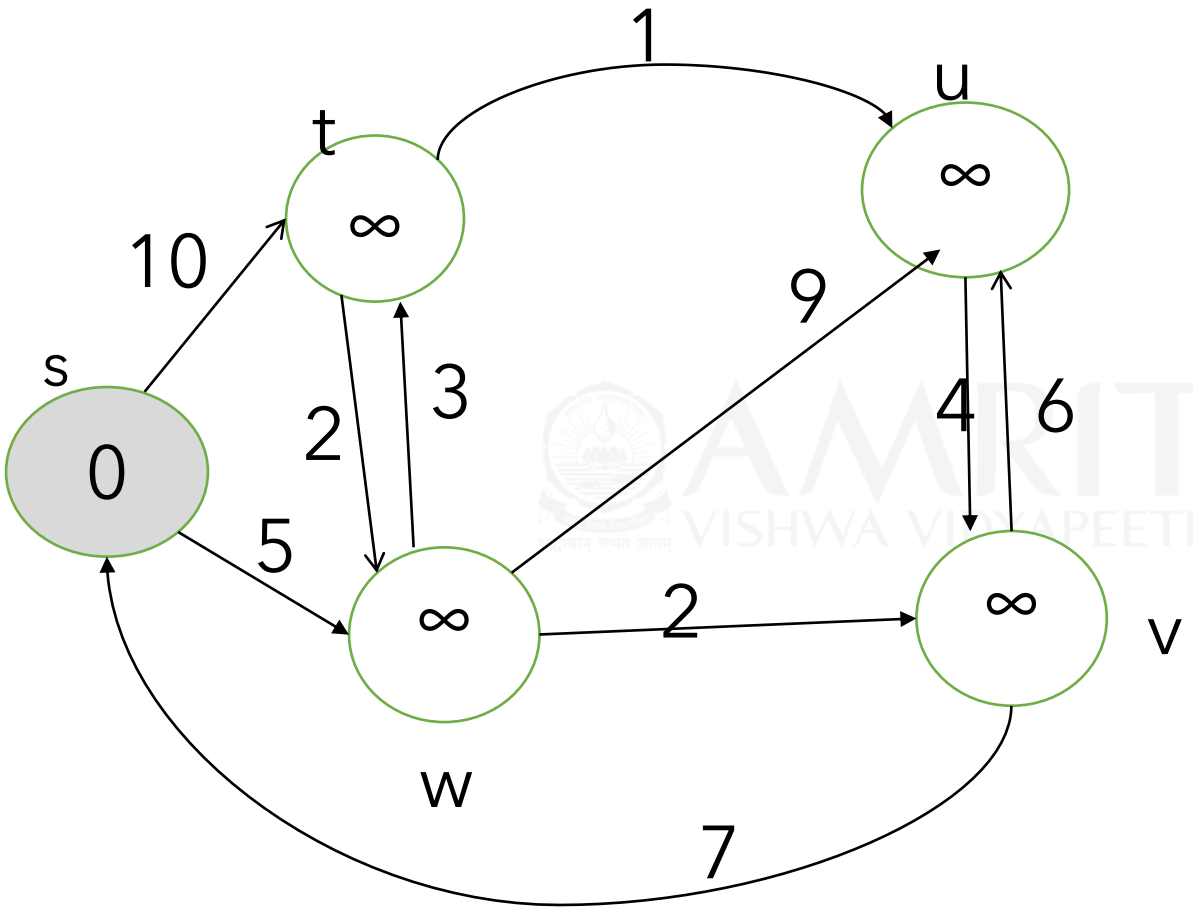
- Single Source Shortest Path.
- Works on weighted directed graph without negative edge weights.
- Running Time of Dijkstra's algorithm is lower than that of the Bellman-Ford algorithm

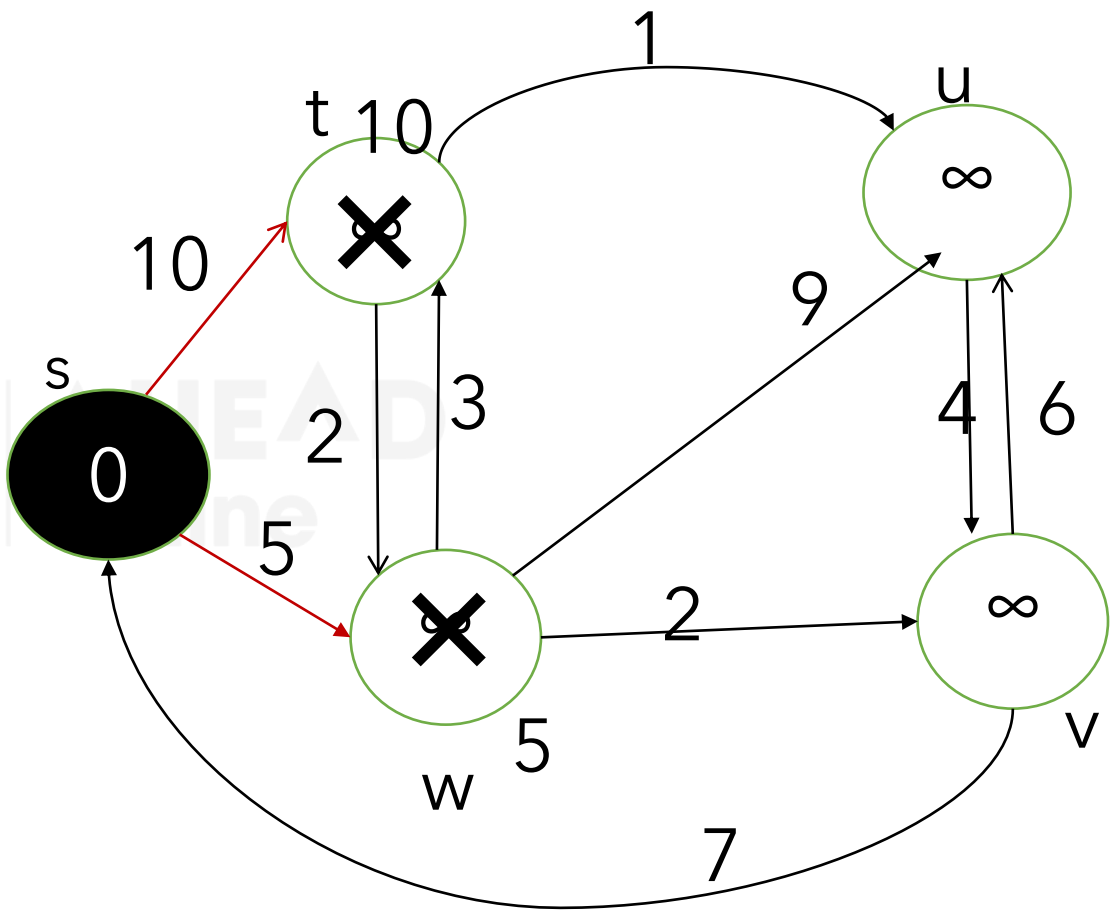
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Dijkstra's algorithm

- Maintains a set S, with final shortest path from source.
- Initialize source s.d=0, each vertex v.d=infinity, v.pred=Nil.
- Algorithm selects the vertex u in V - S with the minimum shortest-path estimate, adds u to S, and relaxes all edges leaving u.
- A min-priority queue Q, that initially contains all vertices of Graph, G =(V,E) is maintained. Vertices are removed from the queue based on distance value.

Dijkstra's algorithm

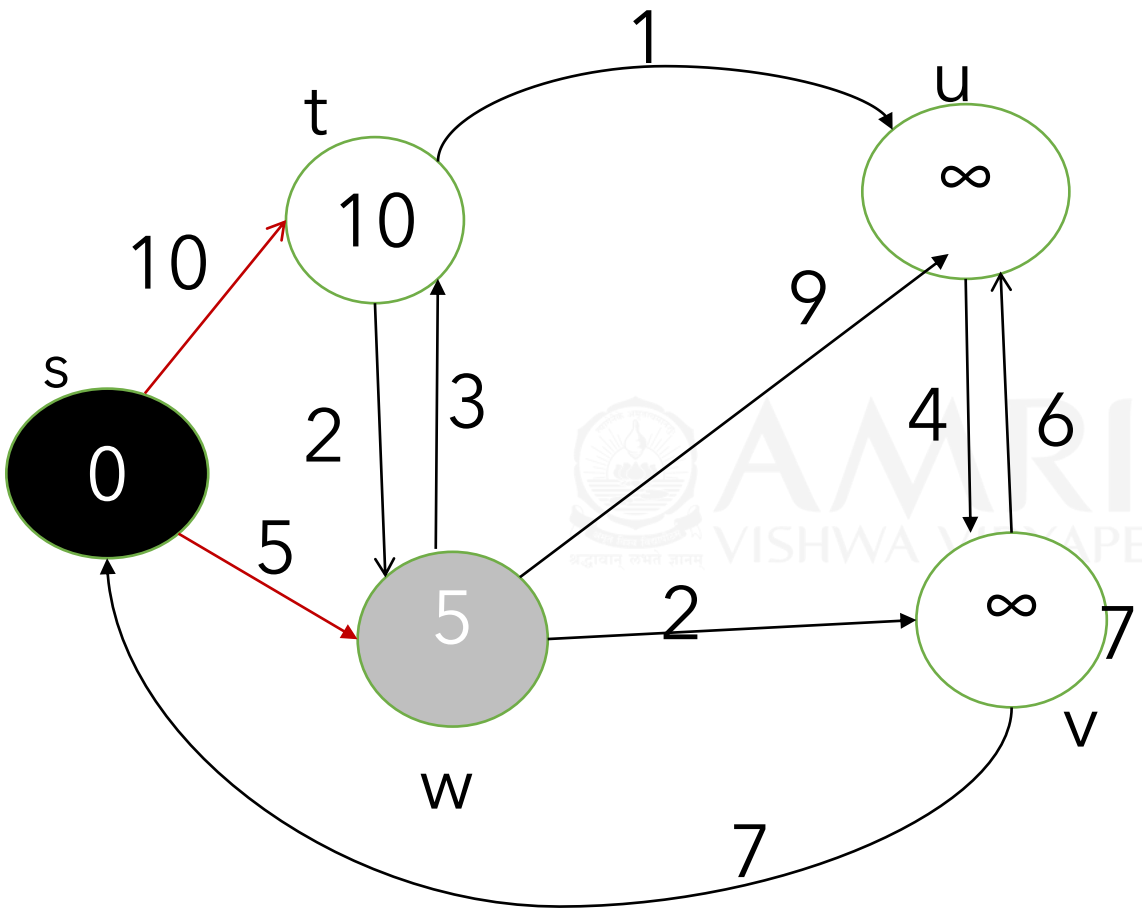


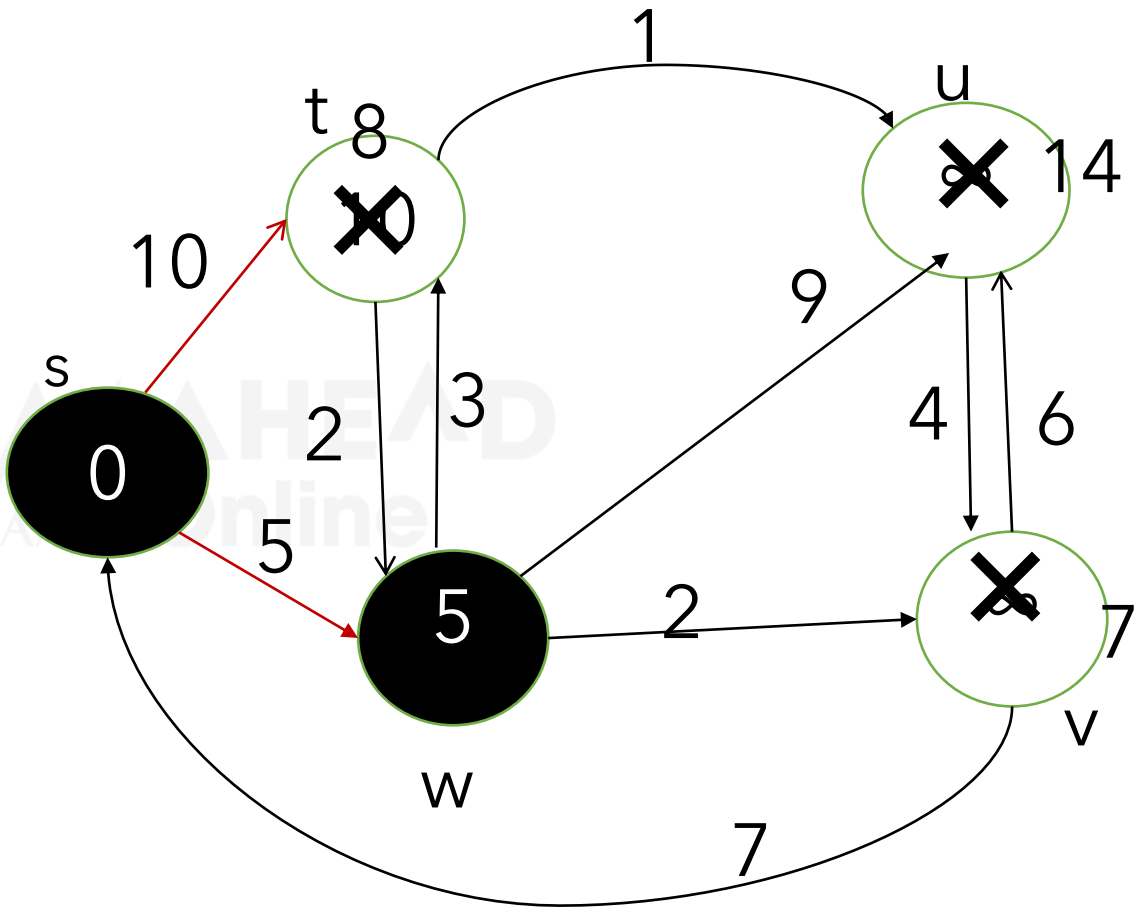


Q= s,t,w,u,v S={}

Q= t,w,u,v S={s}

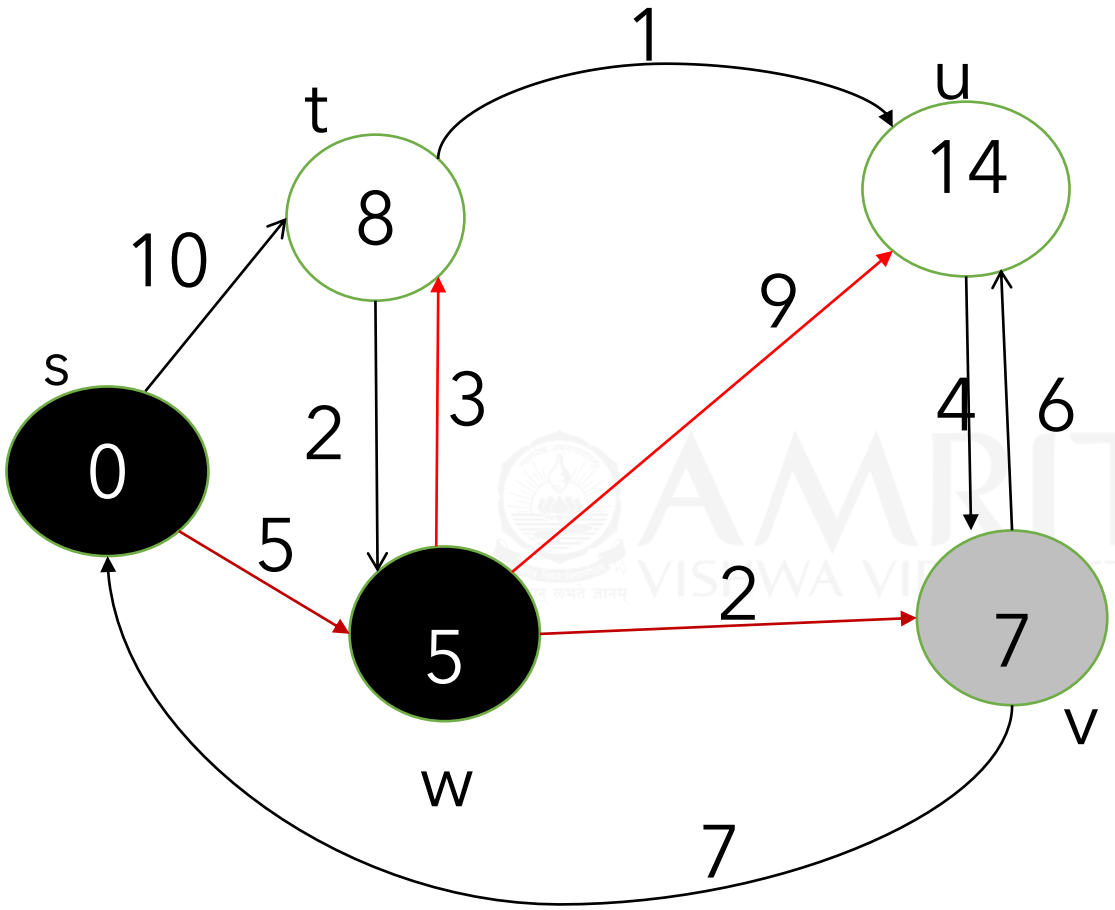
Dijkstra's algorithm



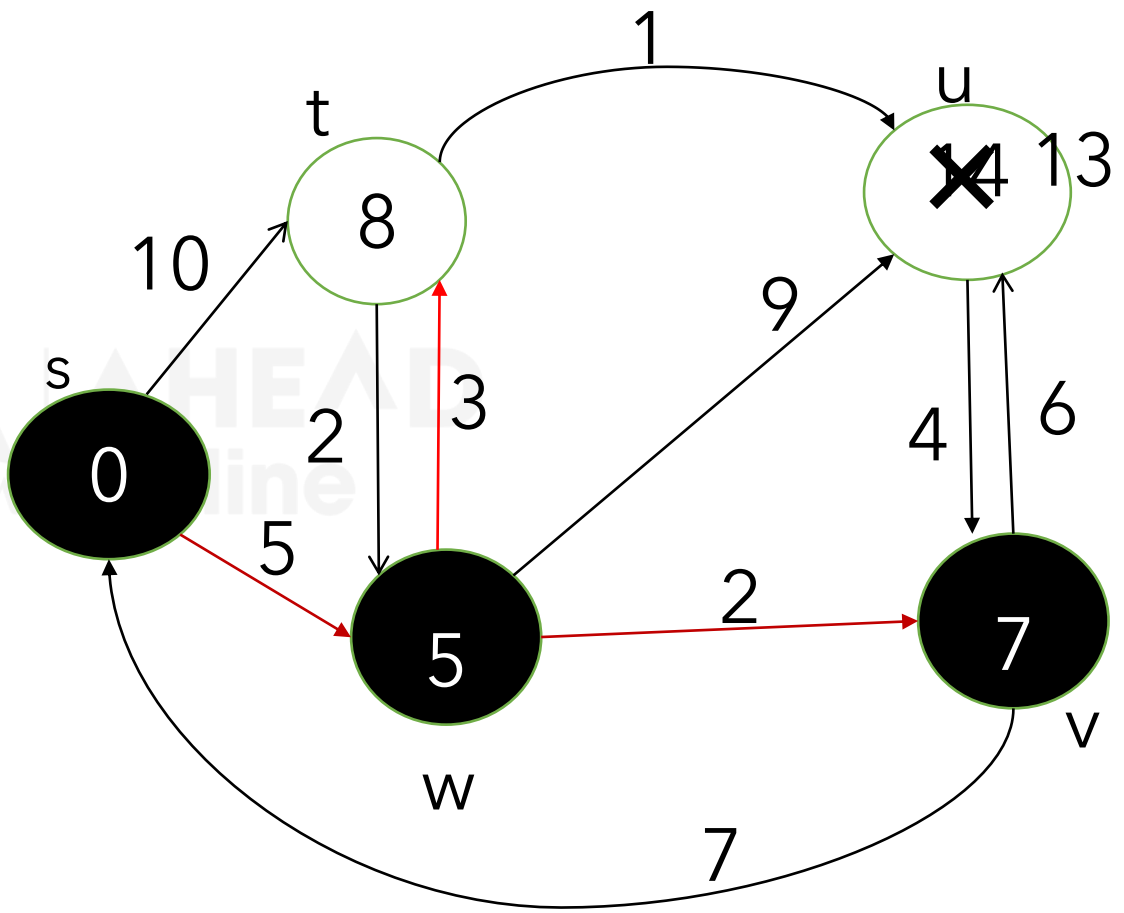


Q= t,u,v S={s,w}

Dijkstra's algorithm

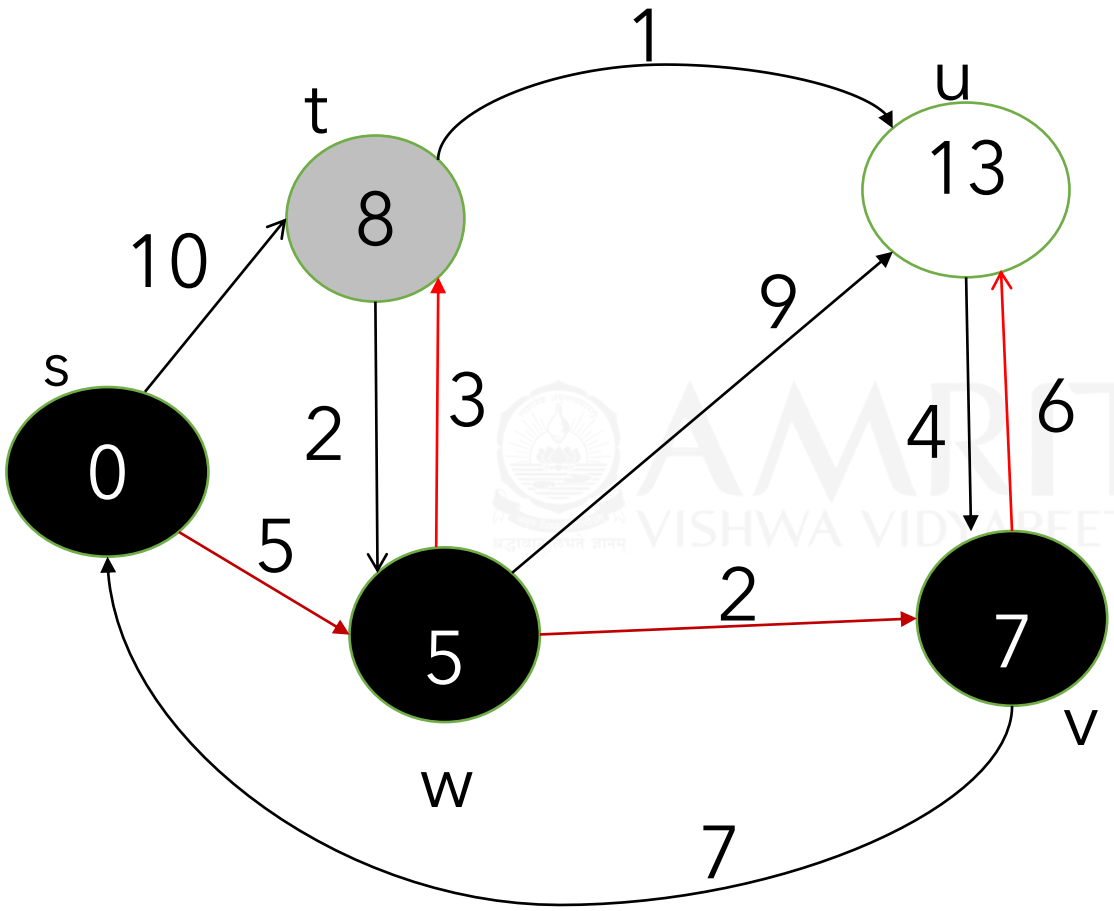


Q= t,u,v S={s,w}

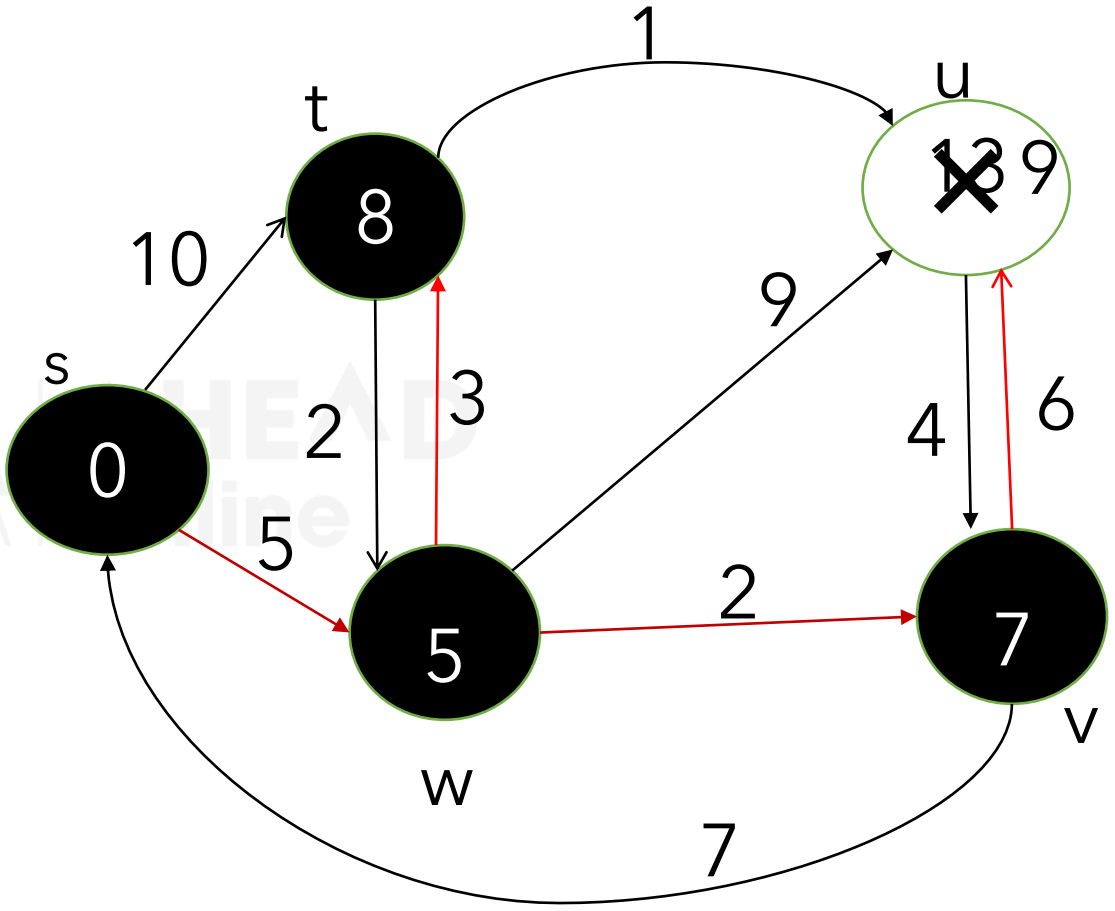


Q= t,u S={s,w,v}

Dijkstra's algorithm

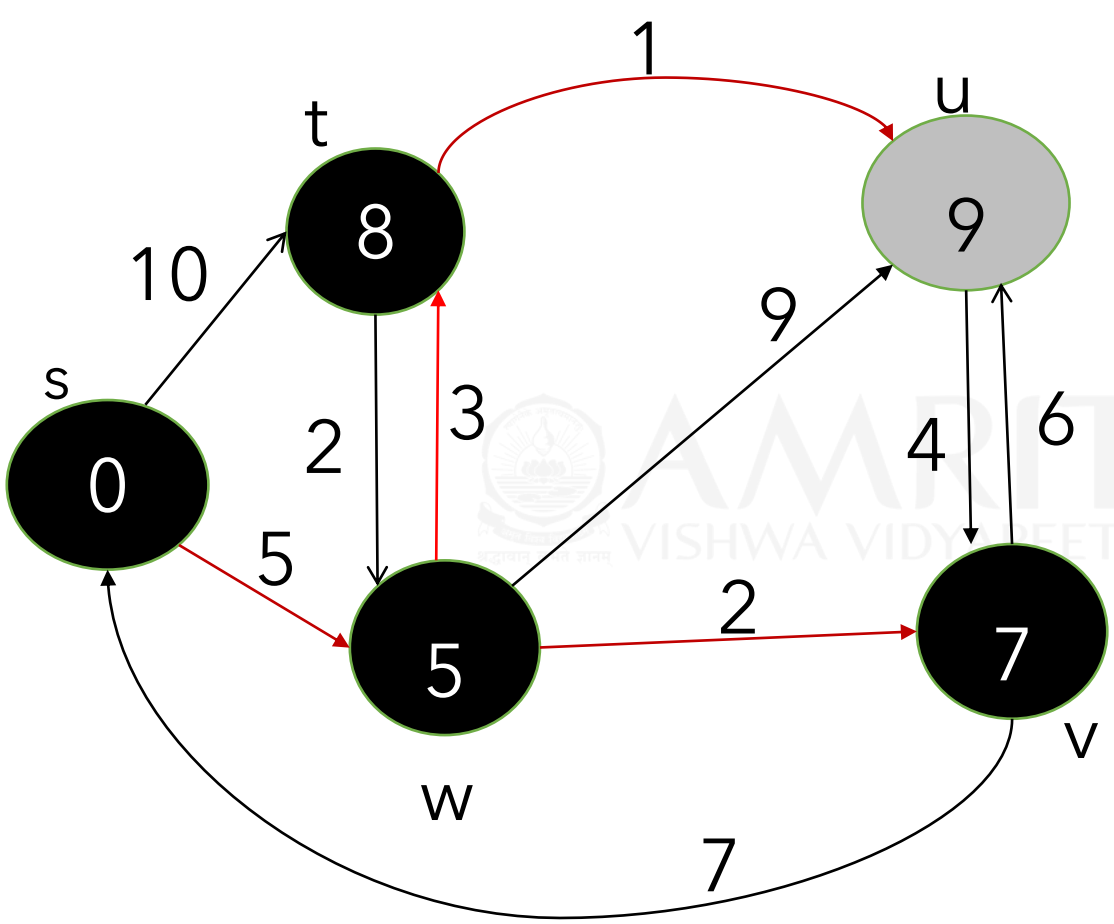


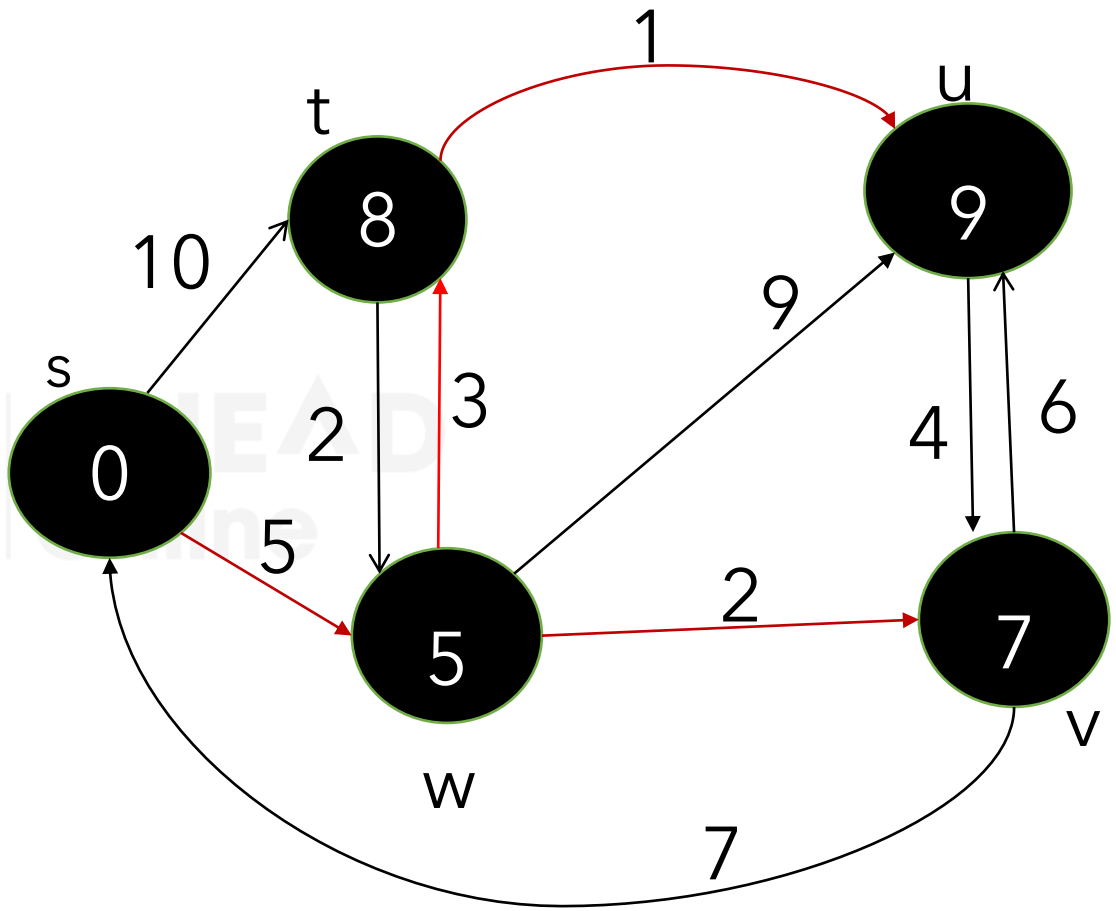
Q= t,u S={s,w,v}



Q= u S={s,w,v,t}

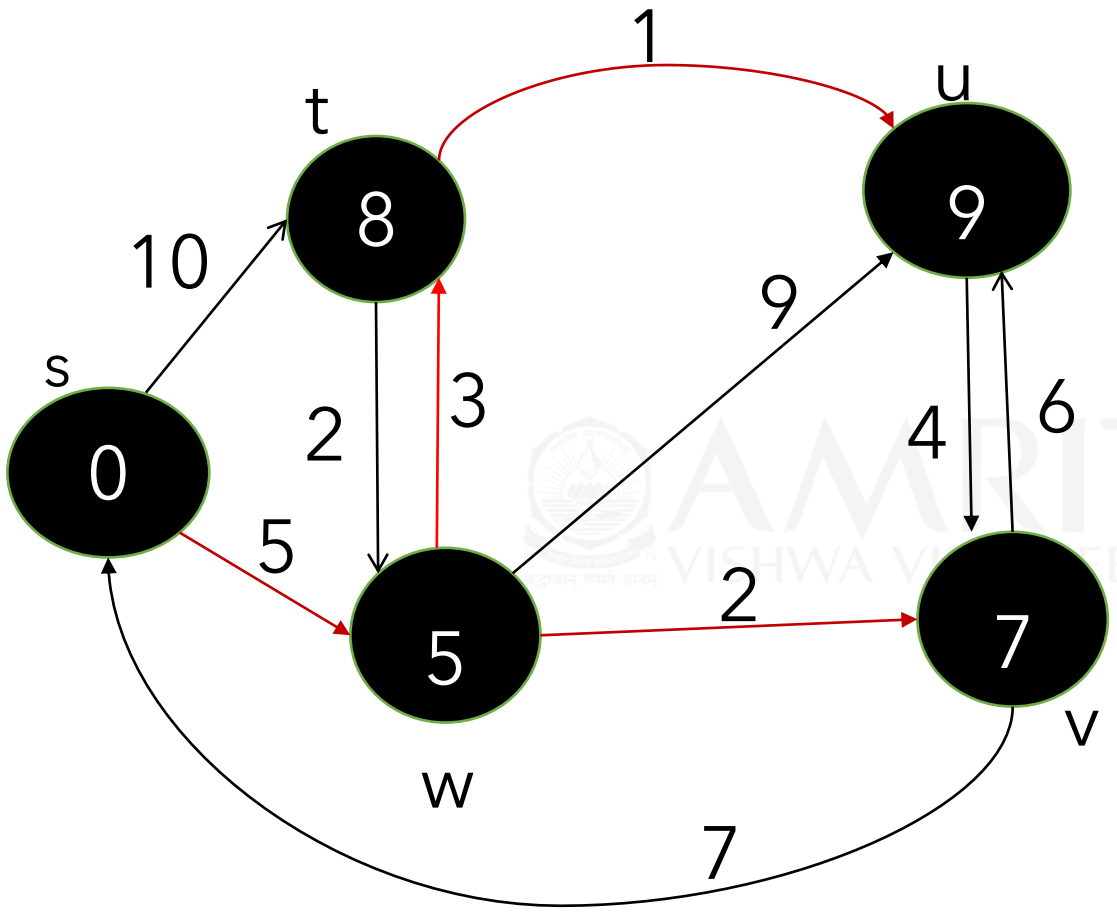
Dijkstra's algorithm





Q= empty S={s,w,v,t,u}

Dijkstra's algorithm



Path to t – s,w,t
Path to w –s,w
Path to u – s,w,t,u
Path to v – s,w,v

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Dijkstra's Algorithm

```
1  Dijkstra(G, w, s):
2      Initialise(G,s)
3      S = ∅
4      Q = G.V
5      while Q ≠ ∅ do
6          u = ExtractMin(Q)
7          S = S U{u}
8          for each vertex v ∈ G.Adj[u] do
9              Relax(u,v,w)
10         end
11     end
12 end
```

```
1 Initialise(G, s):
2     for each vertex v ∈ G.V do
3         v.d = ∞
4         v.pred = NIL
5     end
6     s.d = 0
7 end
```

```
1 Relax(u, v, w):
2     if v.d > u.d + w(u, v) then
3         v.d = u.d + w(u, v)
4         v.pred = u
5     end
6 end
```




Summary

Dijkstra's uses greedy strategy.

When a vertex, u is added to S, u.d is the shortest distance from s to u.

Even though it adopts greedy strategy, it guarantees shortest path computation.

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Reference

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